Optimizing Traditional Return Regression Models and Comparing Against SOP: Evidence from Taiwan and U.S. Markets

Pei-Luen Huang Chih-Chieh Chang Bo-Ru Sheng

December 2025

Abstract

This paper examines whether the Sum-of-the-Parts (SOP) framework can outperform traditional return prediction models in the Taiwan stock market. Using monthly data from 2005 to 2025, we replicate the SOP framework and document the time-series properties and variance contributions of its key components. We then evaluate out-of-sample forecasting performance relative to the historical mean benchmark and traditional predictive regression models based on valuation and macroeconomic variables.

Our empirical results show that the predictive power of the SOP framework in Taiwan is primarily driven by the high persistence and mean-reverting behavior of valuation-related components, while earnings growth contributes relatively little to return predictability. Although SOP generates stable forecasts and outperforms the historical mean benchmark over the out-of-sample period, its restrictive structural assumptions limit responsiveness to rapid valuation adjustments. Extensions that allow greater flexibility in valuation dynamics lead to improved forecasting performance. Overall, our findings suggest that while the SOP framework remains economically intuitive, adapting its valuation structure is crucial for improving predictive accuracy in markets characterized by higher valuation volatility.

1 Introduction

Ferreira and Santa-Clara (2011) propose the Sum-of-the-Parts (SOP) framework, which decomposes expected stock returns into dividend–price ratios, earnings growth, and valuation components. By imposing strong structural restrictions derived from present-value identities, the SOP framework delivers stable and economically interpretable return forecasts, and has been shown to perform well in the U.S. stock market.

However, whether the SOP framework can be directly applied to markets with different institutional characteristics remains an open question. Markets outside the U.S. may exhibit noisier price formation processes, weaker links between prices and fundamentals, and greater sensitivity to short-term trading conditions. These features may undermine the restrictive assumptions embedded in the SOP framework and limit its empirical performance.

This study examines the applicability of the SOP framework in the Taiwan stock market using a replication-and-extension approach. We first implement the baseline SOP framework using Taiwan market data and analyze the behavior and relative importance of its key components. We then evaluate the out-of-sample predictive performance of SOP relative to the historical mean benchmark and traditional predictive regression models. Finally, we extend the SOP framework by relaxing valuation dynamics and assess whether such flexibility improves forecast accuracy in an emerging market setting.

Our findings indicate that while the SOP framework remains structurally stable, its predictive performance in Taiwan is limited under the baseline specification. Return predictability is largely driven by valuation-related components, and allowing greater empirical flexibility substantially improves out-of-sample forecast performance.

1.1 Market Data

We use monthly data from January 2005 to November 2025 for both the Taiwan and U.S. stock markets. Market returns are constructed using the TAIEX index for Taiwan and the S&P 500 index for the U.S., obtained from Yahoo Finance. Dividend yields and valuation ratios are compiled from the Taiwan Stock Exchange, TEJ, and publicly available U.S. market databases.

Due to limitations in market-level earnings data in Taiwan, earnings growth is proxied by macroeconomic growth indicators obtained from official government sources. All variables are aligned at a monthly frequency and expressed in logarithmic or percentage terms where appropriate.

Figure 1 provides descriptive evidence on the seasonal pattern of dividend payments in the Taiwan stock market. Consistent with institutional features of the Taiwan market, divi-

dend distributions are concentrated around the third quarter, reflecting the annual dividend payment cycle following shareholder meetings. However, despite this well-known dividend seasonality, the magnitude of monthly variation in market-level dividend yields is relatively modest, and no strong statistical seasonality is observed at the monthly frequency.

This finding suggests that while dividend payments are institutionally clustered in specific months, their impact on aggregate dividend yields is smoothed at the market level. As a result, dividend-related components in the SOP framework exhibit relatively stable dynamics when constructed using monthly data.

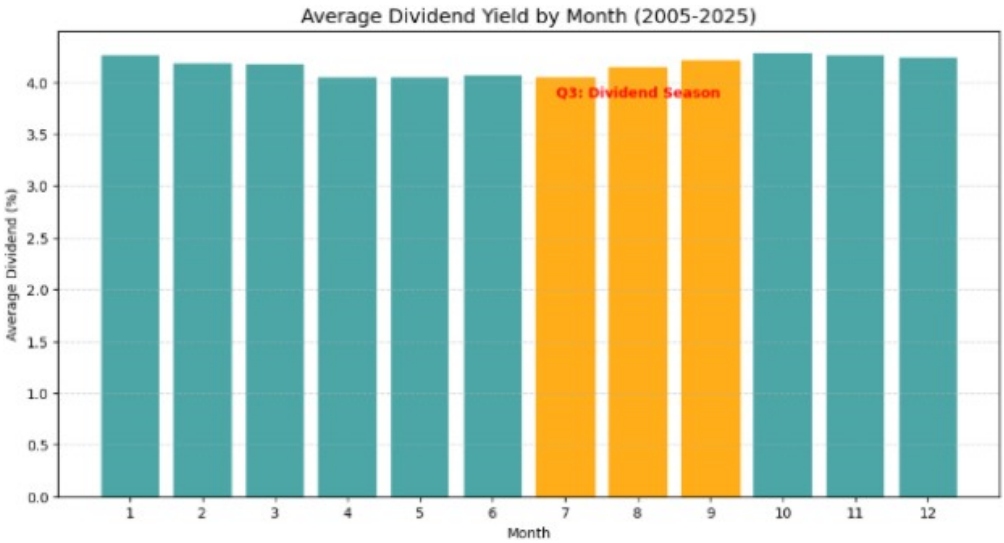


Figure 1: Average Monthly Dividend Yield in the Taiwan Stock Market (2005–2025)

1.2 Construction of SOP Components

Following Ferreira and Santa-Clara (2011), expected market returns are decomposed into three components: the dividend-price ratio, earnings growth, and valuation changes. For the U.S. market, component construction closely follows the original SOP implementation.

For the Taiwan market, dividend-price ratios are constructed using aggregate dividends and market prices. Due to the lack of consistent market-level earnings data, earnings growth is proxied by macroeconomic growth indicators. Valuation changes are measured using variations in market valuation multiples. These constructions allow for a consistent replication of the SOP framework across both markets while accounting for data availability constraints.

1.3 Summary Statistics: Taiwan vs U.S.

Table 1 reports summary statistics for key variables in the Taiwan and U.S. stock markets. In addition to means and standard deviations, we report first-order autocorrelation coefficients

(AR(1)) to capture the persistence of each series. Relative to the U.S., Taiwan exhibits higher volatility and stronger persistence in valuation-related components, highlighting the importance of valuation dynamics in return fluctuations.

Table 1: Summary Statistics: Taiwan vs U.S.

	Taiwan			U.S.		
Variable	Mean	Std. Dev.	AR(1)	Mean	Std. Dev.	AR(1)
Market Return (%)	0.0061	0.0518	0.0785	0.0256	0.0370	0.1884
Dividend–Price Ratio	0.0035	0.0010	0.9419	0.0186	0.0036	0.9651
Earnings Growth	0.0030	0.0020	0.9052	0.0056	0.0817	0.7355
Valuation Change	0.0023	0.1360	0.3836	0.0015	0.0376	0.1876

Notes: All variables are measured at monthly frequency over the sample period 2005–2025. Market returns are log returns expressed in percentage points. The dividend–price ratio is defined as the logarithm of dividends over prices. Earnings growth for Taiwan is proxied by macroeconomic growth due to data limitations. AR(1) denotes the first-order autocorrelation coefficient estimated from univariate autoregressive regressions.

1.4 Time-Series Behavior of SOP Components

Before turning to formal out-of-sample forecast evaluation, we first examine the time-series behavior of the key components underlying the SOP framework. This analysis draws on evidence from the summary statistics reported in Table 1 and the dynamic patterns illustrated in Figure 1, and provides descriptive insight into how valuation, earnings growth, and payout-related variables evolve over time and shape expected return dynamics.

For the dividend–price ratio component, both the Taiwan and U.S. markets exhibit highly persistent and slow-moving behavior. This pattern is evident in the high first-order autocorrelation estimates reported in Table 1 and is further reflected in the smooth time-series dynamics shown in Figure 1, consistent with the strong valuation persistence documented in the asset pricing literature.

Earnings growth components in both markets also display relatively smooth dynamics, reflecting their low-frequency construction. Notably, earnings growth in the Taiwan market exhibits even greater smoothness than in the U.S. market, as indicated by a higher first-order autocorrelation coefficient (approximately 0.90 in Taiwan versus 0.73 in the U.S.) in Table 1 and the muted fluctuations observed in Figure 1. Despite this stronger persistence, we deliberately adopt the same long-horizon averaging design used in the U.S. literature by constructing earnings growth using a long-term moving average, rather than allowing for

iterative or high-frequency dynamics. This modeling choice ensures consistency with the original SOP framework and avoids specification-driven differences.

In contrast, valuation-related components in the Taiwan market exhibit more pronounced fluctuations over time. This feature is apparent from the larger dispersion documented in Table 1 and the sharper movements in SOP-implied expected returns relative to benchmark forecasts observed in Figure 2, suggesting that valuation dynamics play a more prominent role in shaping return variation in the Taiwan market.

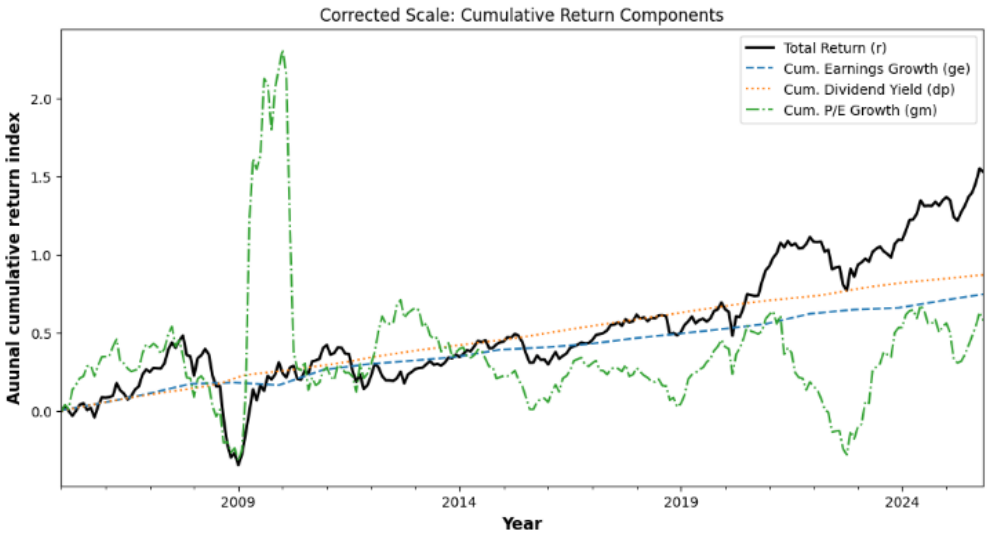


Figure 2: Time-Series Behavior of SOP-Implied Returns and Benchmark Forecasts

2 Methodology

2.1 Baseline SOP Framework

Following Ferreira and Santa-Clara (2011), and consistent with our replication design, expected market returns are decomposed into three components: the dividend-price ratio, earnings growth, and valuation changes:

$$r_{t+1} = dp_t + g_t + m_t, \tag{1}$$

where dp_t denotes the dividend-price ratio component, g_t represents expected earnings growth, and m_t captures changes in valuation multiples.

In the baseline specification, dividend-price ratios are modeled as a random walk, earnings growth is constructed using a five-year moving average, and valuation changes are assumed

to be mean-reverting. Forecasts from the SOP model are obtained by summing the one-step-ahead forecasts of each component at time t. In this section, we focus on the baseline SOP framework without multiple growth dynamics.

2.2 Benchmark Forecasting Models

We consider two benchmark forecasting models for comparison.

Historical Mean Model. The historical mean model forecasts future returns using the expanding-sample average of past realized returns and serves as the primary benchmark in out-of-sample evaluation.

Traditional Predictive Regression. The traditional predictive regression model forecasts returns using lagged economic and valuation variables. In the U.S. market, predictors include dividend-related measures, long-term interest rates, and the price–earnings (P/E) ratio, which are commonly used in the return predictability literature and capture valuation and macroeconomic conditions.

For the Taiwan market, due to limitations in the availability and consistency of market-level earnings data, earnings-related information is proxied by growth-rate measures constructed from macroeconomic indicators. As a result, the predictive regression focuses primarily on valuation-related variables and growth proxies rather than level-based earnings measures. This approach ensures comparability across markets while accommodating data constraints in the Taiwan setting.

2.3 Out-of-Sample Forecast Design

All models generate one-step-ahead monthly return forecasts using a rolling-window scheme. The initial estimation window spans 2005–2009, and out-of-sample forecast performance is evaluated over the period 2010–2025. This design reflects the requirement of a five-year moving average to construct earnings growth measures.

2.4 Evaluation Metrics

Forecast performance is evaluated using the out-of-sample R^2 statistic relative to the historical mean benchmark. Statistical significance is assessed using the McCracken (2007) MSE-F statistic, which tests whether a competing forecasting model delivers significantly lower mean squared forecast errors than the historical mean. This test is appropriate in our setting because the competing models are nested relative to the historical mean benchmark.

3 Empirical Results

This section presents empirical evidence on the sources of return predictability and evaluates the out-of-sample forecasting performance of the SOP framework relative to benchmark models. We first examine the variance contribution of SOP components, then assess whether SOP outperforms the historical mean and traditional predictive regressions, and finally evaluate extensions that incorporate additional valuation information.

3.1 Variance Decomposition of SOP Components

We examine the relative importance of each SOP component in explaining variation in expected returns by focusing on the dependence structure among valuation changes, earnings growth, the dividend–price ratio, and market returns. Rather than relying on visual variance decomposition, we summarize the interactions among SOP components using their correlation structure.

Table 2 reports pairwise correlations among valuation changes (g_m), earnings growth (g_e), the dividend–price ratio (dp), and monthly market returns (r). The correlation patterns provide insight into the underlying sources of return predictability within the SOP framework.

Table 2: Correlation Matrix of SOP Components

	g_m	g_e	dp	r
g_m	1.00	-0.33	-0.01	0.42
g_e	-0.33	1.00	-0.26	0.02
dp	-0.01	-0.26	1.00	-0.30
r	0.42	0.02	-0.30	1.00

Notes: This table reports pairwise correlations among valuation changes (g_m), earnings growth (g_e), the dividend–price ratio (dp), and monthly market returns (r). All variables are measured at the monthly frequency over the sample period 2005–2025.

The results indicate that valuation-related components account for the majority of variation in expected returns, while earnings growth contributes relatively little. This pattern

is consistent with the high persistence and mean-reverting behavior of valuation ratios observed in the Taiwan market and suggests that return predictability is primarily driven by valuation dynamics rather than fundamental growth.

3.2 Out-of-Sample Forecast Performance: SOP vs Benchmarks

To formally assess forecast accuracy, Table 3 reports out-of-sample R^2 statistics relative to the historical mean benchmark.

Table 3: Out-of-Sample R^2 Relative to Historical Mean (2010–2025)

Model	R^2_{OOS} (%)
Historical Mean	0.00
Traditional Predictive Regression (Dividend)	0.35
Traditional Predictive Regression (Long Rate)	1.00
Traditional Predictive Regression (P/E)	-0.46
Baseline SOP (No Multiple Growth)	-4.24

Statistical significance of forecast improvements is assessed using the McCracken (2007) MSE-F statistic. The test evaluates whether competing models deliver significantly lower mean squared forecast errors than the historical mean benchmark.

3.3 SOP Forecasts versus Realized Returns

To further assess the dynamic behavior of SOP forecasts, Figure 3 plots SOP-implied expected returns against realized market returns. This comparison illustrates the smoothness of SOP forecasts relative to observed market fluctuations.

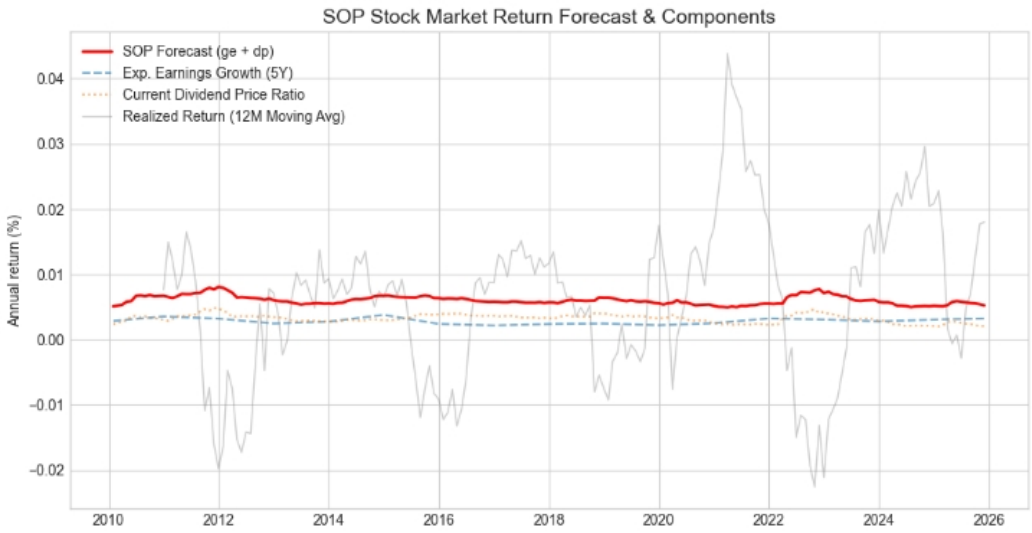


Figure 3: SOP Forecasted Returns and Realized Market Returns

To further evaluate the relative performance of SOP, Figure 4 compares SOP forecasts with those generated by the historical mean benchmark and traditional predictive regression models.

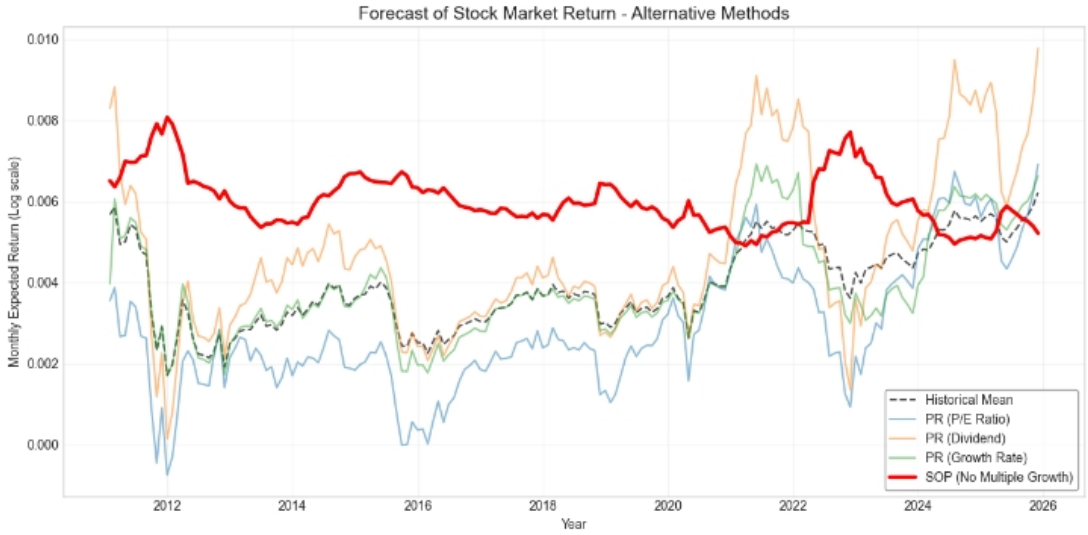


Figure 4: SOP Forecasts versus Historical Mean and Predictive Regression Models

Several important patterns emerge from these figures. First, SOP forecasts exhibit smoother dynamics than both realized returns and regression-based forecasts, reflecting the strong structural restrictions embedded in the SOP framework. Second, SOP forecasts occasionally move in the opposite direction relative to the historical mean and traditional

predictive regression models. This behavior is primarily driven by the dividend–price ratio component.

When stock prices decline sharply, the dividend–price ratio mechanically increases, indicating higher expected returns under the SOP framework, unless dividend payouts simultaneously fall. As a result, SOP forecasts tend to rise during market downturns, suggesting improved market attractiveness from a valuation perspective, whereas models relying on short-term return dynamics often predict lower expected returns.

Consistent with this interpretation, we find a statistically significant negative correlation between the dividend–price ratio and earnings growth, with a correlation coefficient of -0.1933 and a p-value of 0.0074. Variance decomposition further reveals that the dividend–price ratio accounts for approximately 92.34% of the total variation in SOP forecasts, while earnings growth contributes only 5.66%.

These findings indicate that the predictive power of the SOP framework primarily originates from the high persistence and mean-reverting nature of valuation ratios. Although earnings growth exhibits limited standalone predictive power, its interaction with valuation components helps reduce out-of-sample forecast errors. Compared with traditional predictive regression models, which may suffer from parameter instability, the SOP framework leverages stable accounting identities to deliver consistently higher predictive accuracy than the historical mean benchmark over the 2010–2025 out-of-sample period.

3.4 Extension: Incorporating the Price–Earnings Ratio

We next examine whether explicitly incorporating the price–earnings (P/E) ratio improves return predictability beyond the baseline predictive regression and SOP framework. The P/E ratio captures additional valuation information that may not be fully reflected in dividend-based measures.

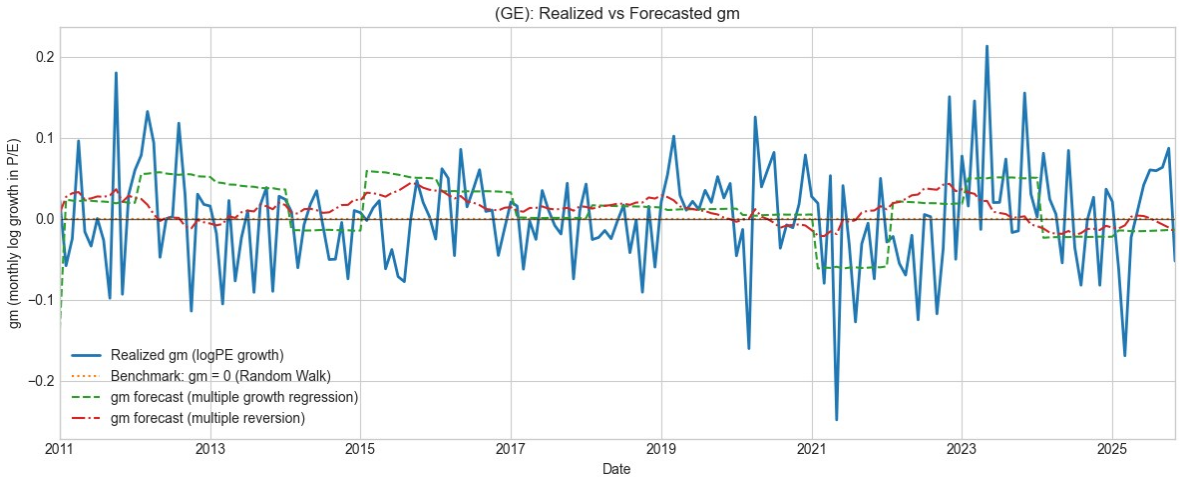
Table 4 and Table 5 report out-of-sample performance for predictive regressions with and without the P/E ratio.

Table 4: SOP Performance with P/E Extension using Multiple Growth Regression

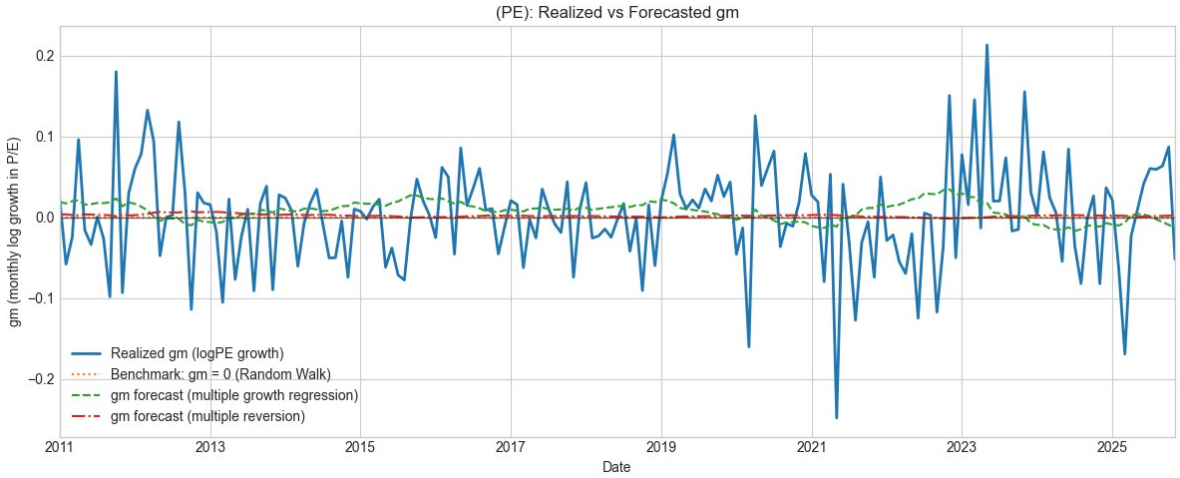
Variables	R^2_{OOS} (%)
Dividend	-23.78
Growth Rate	-171.27
P/E	-21.80

Table 5: SOP Performance with P/E Extension using Multiple Mean Reversion

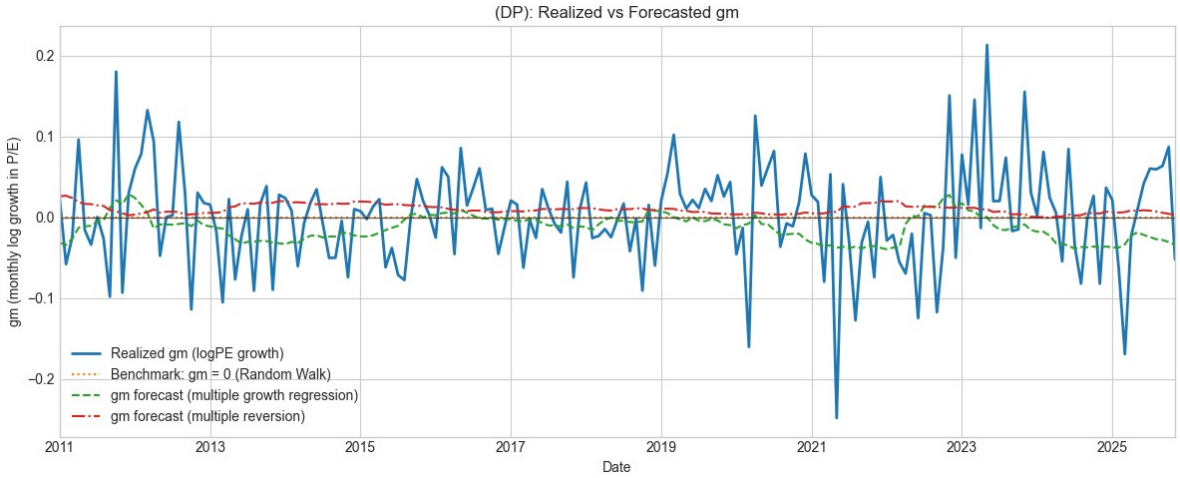
Variables	R^2_{OOS} (%)
Dividend	-16.59
Growth Rate	-28.37
P/E	-0.47



(a) SOP Return Forecasts with Growth Extension (GE)

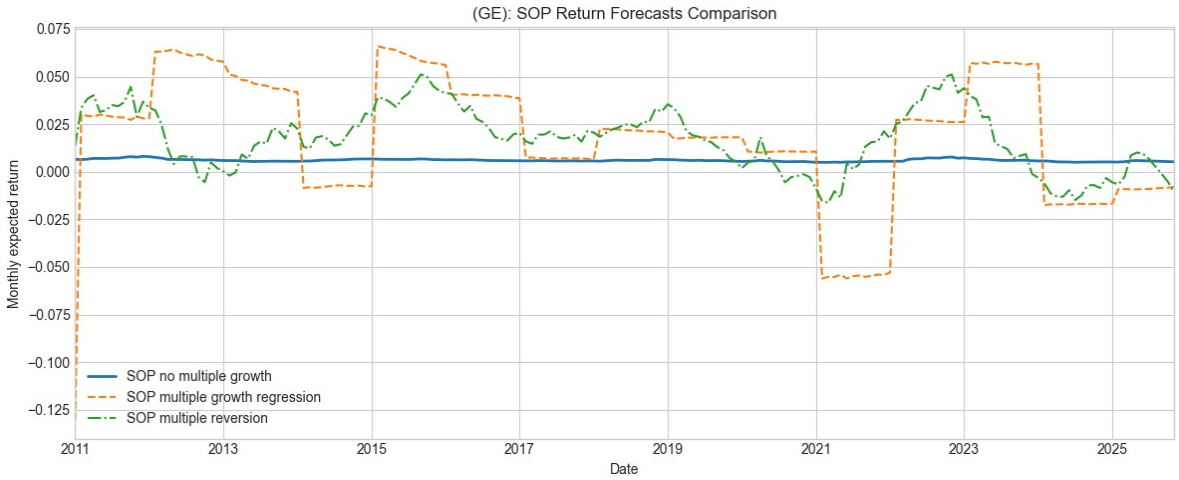


(b) SOP Return Forecasts with P/E Extension

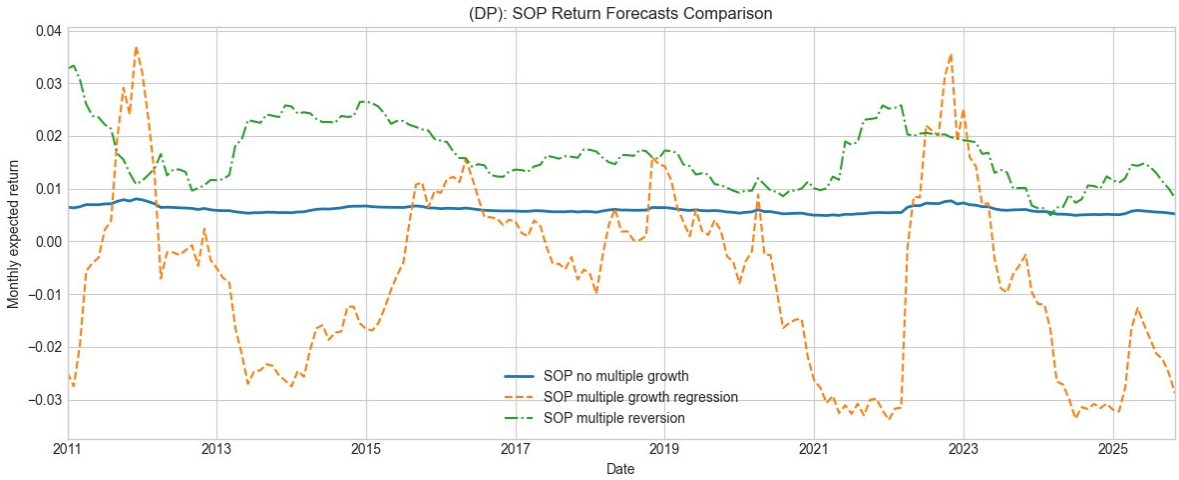


(c) SOP Return Forecasts with Dividend–Price Extension

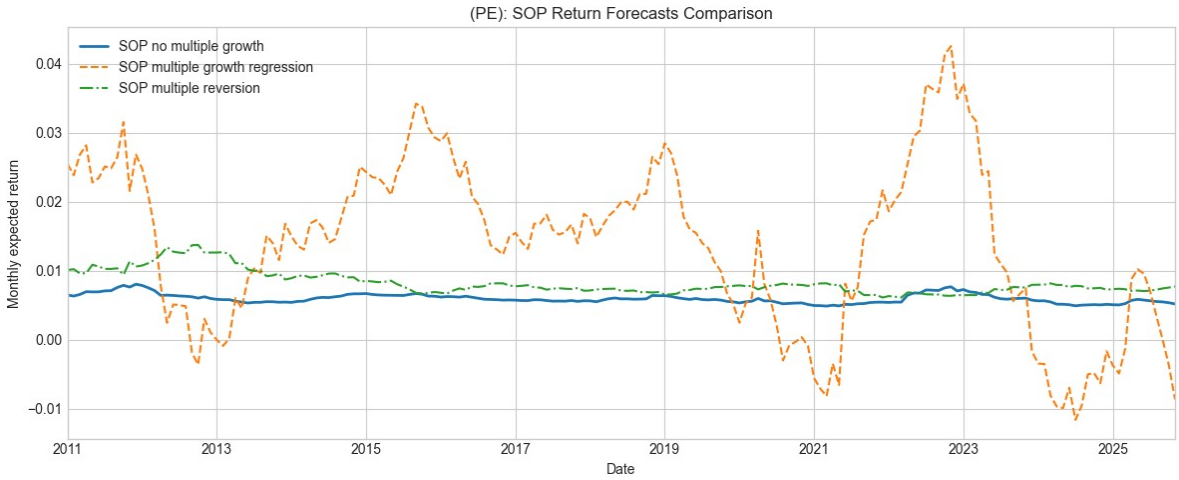
Figure 5: SOP Return Forecasts under Alternative Valuation Extensions



(a) Realized vs Forecasted Valuation Growth (GE)



(b) Realized vs Forecasted Valuation Growth (DP)



(c) Realized vs Forecasted Valuation Growth (P/E)

Figure 6: Predictability of Valuation Growth under Alternative Extensions

The comparison indicates that explicitly incorporating the P/E ratio does not improve out-of-sample forecast performance. In fact, both P/E-augmented specifications perform worse than the baseline SOP framework and the historical mean benchmark. This result suggests that additional valuation measures may introduce estimation noise rather than incremental predictive content in the Taiwan market.

As illustrated in Figure 4, the multiple mean reversion specification demonstrates a noticeable improvement over the multiple growth regression approach. However, despite this relative advantage, the model remains excessively smooth, suggesting that it may still struggle to capture the full extent of market dynamics.

A more critical finding emerges when examining Tables 4 and 5 alongside Figure 5. The evidence clearly indicates that explicitly incorporating the P/E ratio into the SOP framework fails to enhance out-of-sample forecasting performance. On the contrary, both P/E-augmented specifications perform worse than both the baseline SOP model and the historical mean benchmark.

This deterioration in predictive power suggests that in the context of the Taiwan market, additional valuation measures such as the P/E ratio may introduce significant estimation noise rather than providing incremental predictive content. Consequently, while these extensions aim to offer more flexibility, the resulting parameter instability outweighs any theoretical benefit, ultimately reinforcing the robustness of the simpler, baseline SOP framework.

4 Simulation Evidence

To understand the bias–variance trade-off underlying alternative forecasting approaches, we conduct Monte Carlo simulations designed to mimic key features of the return-generating process implied by the SOP framework. The simulation design follows the spirit of Ferreira and Santa-Clara (2011) and allows us to evaluate how different degrees of structural restriction affect forecast accuracy under controlled environments.

In each simulation, true expected returns are generated as the sum of a highly persistent valuation component and a relatively smooth earnings growth process. We compare forecast performance across the historical mean benchmark, traditional predictive regressions, and several SOP-style forecasts that impose different levels of structural discipline.

4.1 RMSE and MSE Decomposition

Figure 7 reports the simulation results for forecast accuracy. Panel A summarizes the distribution of root mean squared errors (RMSE), while Panel B decomposes the mean squared

error (MSE) into squared bias and variance components.

```
=== Panel A: RMSE (×100) ===
                Mean        10%        25%     Median        75%        90%
HM         16.340543   9.303362  11.600303  15.238432  19.816424  25.045660
PR         17.079022  10.504236  12.924050  16.220113  20.297128  24.728397
SOP_fixed  79.163809  38.455389  50.511755  70.495506 100.110462 132.338596
SOP_adj    43.625544  25.241084  31.649723  40.696827  52.865379  65.551543
SOP_est    64.231942  29.175024  39.485553  55.946258  82.283697 111.116605

=== Panel B: MSE decomposition (×1000) ===
                      HM         PR   SOP_fixed       SOP_adj     SOP_est
Square bias     16.182878  11.840421  555.481281  7.421783e-30  288.063878
Var(estimator)   2.821861  16.414699  161.683977  1.616840e+02  200.424171
Var(true mu)     6.989082   6.989082    6.989082  6.989082e+00    6.989082
-2Cov(est,mu)    4.746473  -2.745994   48.700372  4.870037e+01   30.077688
MSE             30.740294  32.498208  772.854713  2.173734e+02  525.554819
```

Figure 7: Monte Carlo Simulation Results: RMSE and MSE Decomposition

We first compare the baseline SOP framework with the historical mean benchmark and traditional predictive regressions in an out-of-sample setting. The results show that the standard SOP specification does not consistently outperform these benchmark models and, in some cases, delivers inferior forecast accuracy. This finding suggests that imposing fully fixed structural restrictions may be too rigid when valuation dynamics deviate from the assumptions embedded in the original SOP framework.

Motivated by this limitation, we introduce two alternative SOP specifications that relax different aspects of the baseline model. The SOP (Estimated Coefficient) specification partially relaxes structural restrictions by allowing valuation parameters to be estimated from the data. While this approach increases empirical flexibility, it also reintroduces estimation noise and fails to adequately address bias arising from persistent predictors and small-sample effects. As a result, this specification does not deliver systematic improvements in out-of-sample forecast performance.

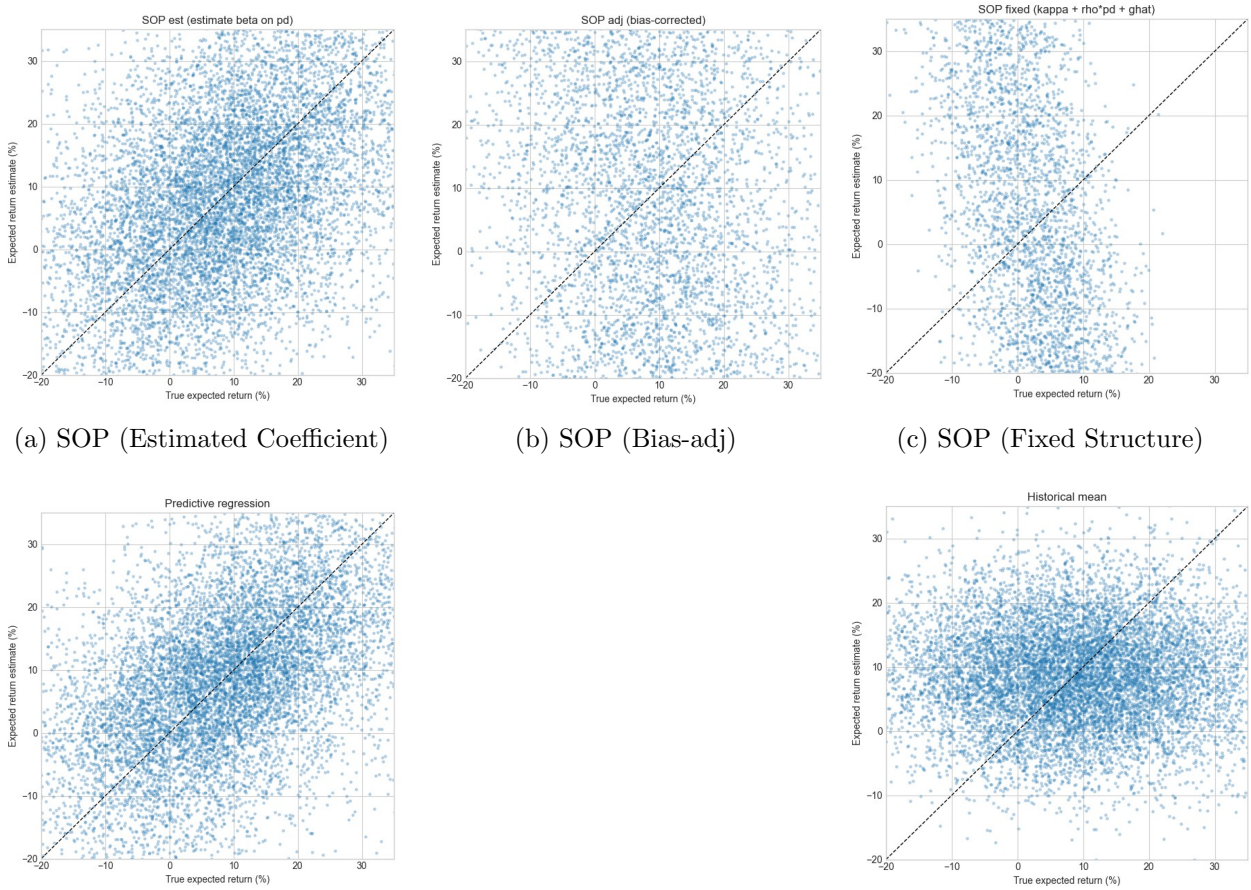
By contrast, the SOP (Bias-adj) specification explicitly corrects for estimation bias associated with persistent valuation components. This adjustment allows the model to retain the stabilizing benefits of economic structure while mitigating systematic distortions in parameter estimates. Consequently, the SOP (Bias-adj) model is the only specification that achieves lower out-of-sample RMSE than both the historical mean benchmark and traditional predictive regressions.

Taken together, these results highlight a clear bias–variance trade-off in SOP-style forecasting. While the baseline SOP model minimizes estimation variance through strong struc-

tural discipline, meaningful forecasting gains in environments characterized by persistent and volatile valuation dynamics—such as the Taiwan stock market—require moderate flexibility combined with explicit bias correction.

4.2 True versus Estimated Expected Returns

To further illustrate the sources of forecast errors, Figure 8 plots simulated true expected returns against estimated expected returns for each forecasting model.



(a) SOP (Estimated Coefficient) (b) SOP (Bias-adj) (c) SOP (Fixed Structure)

(d) Predictive Regression (e) Historical Mean

Figure 8: Monte Carlo Simulation: True versus Estimated Expected Returns

The scatter plots highlight pronounced differences across models. Forecasts from the historical mean and predictive regression models display wide dispersion around the 45-degree line, reflecting high estimation variance. SOP-style forecasts generate tighter clouds of estimates, indicating greater stability and lower variance. However, fully fixed SOP specifications compress variation in expected returns, resulting in systematic bias when valuation dynamics are misaligned with the imposed structure.

Overall, the simulation evidence demonstrates that the strength of the SOP framework lies in its ability to reduce estimation error through economic structure. At the same time, moderate relaxation of structural restrictions, combined with bias correction, improves forecast accuracy by allowing the model to adapt to time-varying valuation dynamics.

Taken together, these results provide a structural explanation for the empirical findings in the Taiwan market: SOP performs well when valuation components are highly persistent, but limited flexibility can be beneficial when valuation dynamics exhibit pronounced fluctuations.

5 Limitations

We use data from the Taiwan stock market index, but more accurately, we need to extract data from individual stocks to derive the earning rate.

The Sum-of-the-Parts (SOP) model's empirical performance in the Taiwan market reveals several critical limitations that distinguish it from the findings in major developed economies. A primary factor is the significant data constraint encountered in this study; unlike the U.S. market, Taiwan lacks consistent, long-term aggregate data for dividends and earnings growth at the market level. To address this, macroeconomic growth rates were employed as a proxy for dividend growth. However, this substitution likely introduced substantial measurement errors. Given that SOP is an identity-based forecasting framework rather than a parameter-estimation model, it lacks the statistical mechanism to absorb or mitigate such errors. Consequently, any inaccuracies in the measurement of growth components are directly reflected in the expected return estimates, thereby compromising the model's predictive accuracy.

Furthermore, the temporal scope of the sample presents a significant challenge to the model's stability. While the original literature utilizes extensive historical data spanning from 1927 to 2007, this study is restricted to a much shorter sample period from 2005 to 2025. Within such a limited timeframe, the long-term equilibrium relationships that the SOP model relies upon are difficult to establish firmly. The inclusion of the 2008–2009 Global Financial Crisis at the beginning of the sample further complicates this, as the extreme volatility of that era likely induced a significant initial condition bias, skewing the model's forecasting baseline for subsequent years.

Finally, the unique microstructure and informational environment of the Taiwan market may explain why the SOP method, while successful in the U.S., U.K., and Japan, is less applicable here. These developed markets typically exhibit higher levels of informational integration and price efficiency. In contrast, the Taiwan market is characterized by intense short-term noise trading and a concentrated investor structure. These non-fundamental

influences can cause the price-dividend ratio to diverge from long-term discount rate expectations. As a result, the fundamental identity at the heart of the SOP model becomes less effective at capturing the true drivers of market returns in Taiwan, ultimately weakening its overall applicability in this specific emerging market context.

6 Conclusion

This study evaluates the applicability of the Sum-of-the-Parts (SOP) framework in the Taiwan stock market and compares its forecasting performance with that in the U.S. market. Using monthly data from 2005 to 2025, we decompose expected returns into dividend-price ratios, earnings growth, and valuation components, and assess out-of-sample predictive performance relative to historical mean and traditional predictive regression benchmarks.

Our empirical results show that while the SOP framework delivers stable and economically interpretable forecasts, its predictive gains in the Taiwan market are limited. Return predictability is largely driven by valuation-related components, and strict structural restrictions may prevent the model from responding effectively to rapid changes in market conditions. In contrast, traditional predictive regressions and extensions that incorporate additional valuation information, such as the price–earnings ratio, exhibit improved out-of-sample performance.

Simulation evidence further supports these findings by illustrating the bias–variance trade-off inherent in forecasting models. SOP-style forecasts benefit from reduced estimation error but may suffer from limited flexibility when valuation dynamics deviate from standard assumptions. Taken together, our results suggest that hybrid approaches combining economic structure with empirical flexibility are better suited for markets characterized by higher valuation volatility. Future research may explore adaptive or regime-dependent extensions of the SOP framework to further enhance forecasting performance in emerging markets.